

The Golden Ratio in FFGFT

Notation, Roles and Status – a Consolidating Clarification

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Abstract

The golden ratio $\varphi = (1 + \sqrt{5})/2$ appears in the FFGFT corpus in several clearly distinguishable roles – and under two different glyphs that partly collide with other quantities (scalar field, angle, potential, wave function). This document does *not* change the earlier texts; it fixes the canonical notation and reading, assigns each substantial occurrence to a role, and cleanly separates what is *derived* from φ , what is merely *asserted*, and what is simply *miscalculated*. Key statements: (i) φ is nowhere a foundation in FFGFT – that is $\xi = 4/30000$; (ii) the only φ usage with genuine geometric content is $\arctan(\varphi^{-3})$ in the CP phase (Doc. 172), because $\varphi^{-3} = \sqrt{5} - 2$ is the natural intermediate scale of Fibonacci convergence; (iii) φ mass ladders (Doc. 172/188) are heuristic, numerically erroneous and fall under the φ^N triviality warning P35; (iv) $1/\varphi$ is a transit point, not a fixed point (Doc. 275); (v) π and e are nowhere derived.

.1 Occasion and Scope

While maintaining the corpus a notation problem became apparent: the same symbol `\phi` means different things in different documents, and the golden ratio itself is written sometimes as ϕ (closed form, from `\phi`), sometimes as φ (open form, from `\varphi`). Since the old documents are finalised in version v1.1.2 and are no longer rewritten retroactively, this document serves as a *binding reference*: anyone who encounters φ or ϕ in an older text looks up here which role is meant.

Important

Scope rule. This document is normative for the *reading*, not for the *wording* of the old documents. Where an old text and this document contradict each other, the version recorded here – reconciled against Doc. 190 (P35/P37) and Doc. 275 – governs the physical content.

.2 Canonical Notation

The following assignment is the canonical convention. It also describes *how* the corpus actually uses the symbols – with the known inconsistency for the golden ratio.

Command	Glyph	Canonical meaning	Remark
<code>\varphi</code>	φ (open)	golden ratio $(1 + \sqrt{5})/2$	recommended standard symbol
<code>\phi</code>	ϕ (closed)	scalar field / angle / index	NOT the golden ratio
<code>\Phi</code>	Φ (capital)	potential / complex field	e.g. $\Phi = \rho e^{i\theta}$
<code>\psi</code>	ψ	wave function	quantum mechanics

Caution

Known legacy issue. In v1.1.2 several documents write the *golden ratio* erroneously as `\phi` instead of `\varphi` (among others Doc. 005, 009, 022, 023, 028, 035, 036, 083, 133, 147, 187, 188). At the same time, field-theory documents use `\phi` correctly for the *scalar field* (e.g. Doc. 202: $\phi_{n,l,j}$ with $\square \phi_{n,l,j} = -m^2 \phi_{n,l,j}$) and `\phi` as a *winding index* n_ϕ in (n_θ, n_ϕ) (e.g. Doc. 051, 202). Within a *single* document ϕ is unambiguous in each case; the overloading exists only *between* documents. Hence one must not replace `\phi`→`\varphi` wholesale – that would destroy the field/angle documents.

Numerical convention. Throughout,

$$\varphi = \frac{1 + \sqrt{5}}{2} = 1.6180339887 \dots, \quad \varphi^2 = \varphi + 1, \quad \varphi^{-1} = \varphi - 1 = 0.6180 \dots, \quad (1)$$

$$\varphi^{-3} = \sqrt{5} - 2 = 0.2360679 \dots, \quad \arctan(\varphi^{-3}) = 13.2825^\circ. \quad (2)$$

.3 The Roles of φ in the Corpus

Building on the threefold division in Doc. 190 (P37) – (1) microscopic/ ξ level, (2) macroscopic/ $1/\varphi$ transit point, (3) particle physics – a finer resolution is given here. Each role is provided with its sources and its status.

Role A – Self-similarity attractor (asserted)

Source: Doc. 172, § “The golden ratio as a fractal attractor”. **Statement:** The fractal field geometry with $D_f = 3 - \xi$ is said to force a self-similarity recursion whose only stable fixed point is φ (algebraically $\varphi^2 = \varphi + 1$). **Status: asserted, not derived.** $\varphi^2 = \varphi + 1$ is the defining equation of φ ; the actual step “ $D_f = 3 - \xi$ forces exactly this recursion” is asserted, not shown. The variant in Doc. 188 (“ φ as the most stable incommensurable ratio”) is more strongly motivated – via the continued fraction $[1; 1, 1, \dots]$ φ is the “most irrational” number – but likewise hinges on the geometry selecting precisely this ratio.

Role B – CP phase $\arctan(\varphi^{-3})$ (genuine content)

Source: Doc. 172, § “Formal proof: $\arctan(\varphi^{-3})$ as a geometric necessity”. **Statement:** The Fibonacci quotients $F(n+1)/F(n)$ converge to φ with errors of order $\varphi^{-2n}/\sqrt{5}$; the geometric mean of first and second order is $\sqrt{\varphi^{-2}\varphi^{-4}} = \varphi^{-3} = \sqrt{5} - 2$. With the torsion phase it follows that $\delta = 270^\circ + \arctan(\varphi^{-3}) \approx 283.28^\circ$ (NOvA/T2K 2024: $283.1^\circ \pm 14^\circ$).

Correct

Status: the best-justified φ usage in the corpus. Here φ^{-3} is not a choice but the natural intermediate scale of Fibonacci convergence; the exponent 3 is forward-motivated via the Z3 order (3 generations/sectors). This is qualitatively different from a fitted φ power.

Role C – Fine-structure constant $\alpha = \xi \varphi^3 F(7)$

Source: Doc. 172, § “ α from pure geometry”. **Statement:**

$$\alpha \approx \xi \cdot \varphi^3 \cdot F(7) = \xi \cdot \varphi^3 \cdot 13 \Rightarrow \alpha^{-1} = 136.19 \quad (-0.62\%). \quad (3)$$

Status: numerically verified as correct, but to be understood as one of several α routes. The main route runs via $\alpha = \xi E_0^2$ with $E_0 = \sqrt{m_e m_\mu / K_{\text{frak}}} = 7.398 \text{ MeV}$ (Doc. 011); its agreement with the measured value is a property of the anchor E_0 , not an independent prediction. The $\varphi^3 \cdot 13$ form is the geometrically “clean” variant, but it deviates from the measured value by 0.62%.

Role D – Mass/generation ladder (heuristic, erroneous)

Sources: Doc. 172 ($m_\mu/m_e \approx \varphi^{10}$), Doc. 188 (lepton ladder $\varphi^5 \pi^2, \varphi^5$), Doc. 009 ($\varphi^{12}/(1-\xi)^2$), Doc. 022/035/133 (φ^{gen} scaling). **Status:** heuristic; in Doc. 172/188 numerically wrong (see Section .4). The robust mass derivation runs *not* via φ but via ξ and quantum numbers (Doc. 006, Section .4). This role falls under P35.

Role E – Fractal scaling factor $\alpha = \varphi^{-1}$ (NOT an error)

Source: Doc. 188, § “Fractal coupling constant”. **Statement:** $\alpha = \varphi^{-1} = 0.6180$ as a scaling factor between fractal levels, $(x_{k+1}, \dots) = \varphi^{-1}(x_k, \dots)$.

Important

Status: its own, clearly defined quantity – not the fine-structure constant. In the same document the fine structure stands cleanly separated as $\alpha_{\text{EM}} = \xi E_0^2 = 1/137.04$. Thus there is *no* computational error, but a deliberate double assignment of the letter α . A pure readability matter; Doc. 190 P37(a) already lists this usage.

Role F – Transit point $1/\varphi$ of the ξ recursion

Source: Doc. 275. **Statement:** The damping sequence $r(k+1) = r(k)(1-\xi)$, $r(0) = D_f/3$, passes $1/\varphi$ at $k^* = \log \varphi/\xi \approx 3609$ (exactly $k_{\text{exact}}^* = 3608.51$).

Caution

Status: transit point, not a fixed point. The only fixed point of the recursion is 0. Since $k^*(c)$ exists for *any* target c , the observation is unanchored as long as $1/\varphi$ is not independently motivated – the φ^N triviality warning P35 applies.

Role G – Pentagonal symmetry breaking

Sources: Doc. 018 (anomalous $g-2$), Doc. 149 (torsion), Doc. 019, 153, 154. **Statement:** φ as the carrier of pentagonal (fivefold) symmetry breaking. **Status:** structural motif; it shares the justification status of Role A (the selection of precisely the pentagonal symmetry is the open point).

Role H – HLV substrate (not FFGFT)

Source: Doc. 271. **Statement:** In Hidden-Layer Vacuum (HLV, Krueger) φ is the substrate geometry. **Status:** belongs to HLV, not FFGFT. In the bridge comparison φ must not be tacitly transferred from the HLV side to the FFGFT side.

Role I – Trivial representation $X = \varphi^N$

Sources: Doc. 190 (P35), 271, 272, 274. **Statement and status:** any magnitude can be written as φ^N or ξ^N ; an exponent alone is not a prediction. Holds equally for both bases.

.4 Honest register: correct / open / wrong

Correctly recomputed

- $\alpha = \xi \varphi^3 \cdot 13 \Rightarrow \alpha^{-1} = 136.19$ (-0.62%), Doc. 172.
- $\varphi^{-3} = \sqrt{5} - 2 = 0.23607$, $\arctan(\varphi^{-3}) = 13.28^\circ$, Doc. 172.
- $1/\varphi = 0.6180$ as a transit point at $k^* \approx 3609$, $r(3609) = 0.617994\dots$, Doc. 275 (status: unanchored).
- $245 = 5 \cdot 7^2 \approx \varphi^{12} / [1.314(1 - \xi)^2] = 245.1$, Doc. 009 – with a *declared* empirical anchor 1.314 (P10/P37(d)).

Numerically wrong – in the legacy documents, corrected here

Critical

Doc. 188, lepton ladder (line ≈ 318):

$$\text{claimed: } \frac{m_\mu}{m_e} \approx \varphi^5 \cdot \pi^2 \approx 206.7 \quad \text{actually } \varphi^5 \pi^2 = 109.46 \text{ (measured 206.768),} \quad (4)$$

$$\text{claimed: } \frac{m_\tau}{m_\mu} \approx \varphi^5 \approx 16.94 \quad \text{actually } \varphi^5 = 11.09 \text{ (measured 16.82).} \quad (5)$$

Doc. 172 (line ≈ 356): $m_\mu/m_e \approx \varphi^{10} = 122.99$ – 40% off and contradicting the (likewise wrong) Doc. 188 form. Even arithmetically corrected one would need awkward exponents ($\varphi^{11.08}$ resp. $\varphi^{5.87}$): there is no clean φ power that hits the lepton ratios.

Consequence: The φ lepton ladder is to be flagged as numerology (P35 null result), not as a geometric consequence.

The robust lepton-mass derivation – via ξ , not φ

Doc. 006 sets $m_i = r_i \xi^{p_i} v$ with exponents fixed from quantum numbers:

$$p_e = \frac{3}{2}, \quad r_e \approx 1.349; \quad p_\mu = 1, \quad r_\mu = \frac{16}{5} = 3.2; \quad (6)$$

$$\frac{m_\mu}{m_e} = \frac{r_\mu}{r_e} \xi^{p_\mu - p_e} = 2.372 \cdot \xi^{-1/2} = 2.372 \cdot 86.6 = 205.4 \quad (-0.65\%). \quad (7)$$

The *exponent difference* $-\frac{1}{2}$ is the structural, forward-directed content; the $O(1)$ coefficients (r_e is back-computed) carry the residual fit (cf. P40: only ratios exact). φ is not needed here and adds nothing.

.5 What does NOT follow from φ (or ξ)

- **π is not derived.** π is the presupposed circle/torus constant. The nicest interpretation (Doc. 159): on the circle $\sum 1/n^2 = \pi^2/6$ converges, the periodic geometry forces convergence – but π is interpreted, not derived.
- **e (Euler's number) is not derived**, it appears only as $\exp(\cdot)$.
- **Emergently derived**, by contrast, are: $13 = F(7)$ (Fibonacci) and $27 = 3^3$ (Z3 structure, three generations).

.6 Table of occurrences (substantial cases)

The table lists the *content-relevant* occurrences. Pure definitional or pass-through mentions follow the same canonical reading.

Doc.	Symbol	Role (see above)	Status
172	φ	A attractor, B CP phase, C α , D φ^{10}	B correct; A asserted; D wrong
188	ϕ	D lepton ladder, E scaling factor	D wrong; E ok (own quantity)
275	φ	F transit point $1/\varphi$	correct, but unanchored (P35)
190	φ	I triviality P35, role map P37	normative
009	ϕ	D $\varphi^{12}/1.314$	empirical anchor declared
271	φ	H HLV substrate, I triviality	HLV/FFGFT delineation
272	φ	I $X = \varphi^N$ not a prediction	correct
274	φ	I triviality holds for φ and ξ	correct
018	φ	G pentagonal symmetry breaking ($g-2$)	structural motif
149	φ	G pentagonal symmetry breaking (torsion)	structural motif
019, 153, 154	φ	G pentagonal symmetry breaking	structural motif
022, 035, 133	ϕ	D φ^{gen} scaling	heuristic (P35)
036	ϕ	D cosmic scales $r_n = r_0 \varphi^n$ (BAO)	heuristic (P35)
083, 147, 187	ϕ	D qubit/photonics frequencies	heuristic
005, 028, 035, 150	ϕ/φ	definition $(1 + \sqrt{5})/2$	basis
241, 261	φ	F “settled” recursion	cf. Doc. 275
202	ϕ	<i>scalar field</i> $\phi_{n,l,j}$, index n_ϕ	NOT golden ratio
051	ϕ	<i>winding index</i> n_ϕ	NOT golden ratio
067, 097, 129	ϕ	<i>scalar field</i> (Lagrangian/QFT)	NOT golden ratio

.7 Recommendation

1. **Notation going forward:** golden ratio exclusively φ . ϕ remains field/angle, Φ potential, ψ wave function.
2. **Legacy documents:** do not rewrite; this reference governs.
3. **φ mass ladders (Doc. 172/188):** read as P35 numerology, not as derivation. Only the ξ route is robust (Doc. 006).
4. **Genuine φ content** remains: $\arctan(\varphi^{-3})$ in the CP phase (Doc. 172) and the conceptual separation of the three φ roles (Doc. 190 P37).